



PRAVEEN KUMAR OJHA

Process & Data Analyst

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MOTIVATION

Process & Data Analyst with 4+ years of specialized experience in process mining and data engineering at Bosch. I excel at bridging the gap between complex data analysis and practical business solutions, with proven expertise in optimizing Celonis platforms and building scalable automation tools. Currently pursuing my Master's thesis on LLM-based process optimization, I'm seeking roles where I can leverage **advanced analytics** to solve **complex operational challenges** and **drive measurable business impact**.

Let's connect if you're solving **complex process challenges** or need **scalable data analysis expertise**.

CORE COMPETENCIES

TECHNICAL SKILLS

Process Mining: Celonis, pycelonis, pm4py, OCPM, Action Flow, ETL

ML & Viz: Python, scikit-learn, tensorflow, seaborn, SQL

Infra & Web: Flask, Docker, Streamlit, Git, Prompt Engineering

DOMAIN EXPERTISE

O2C (Order-to-Cash)

B2C (Bill-to-Cash)

P2P (Procure-to-Pay)

A2A (Affiliate-to-Business)

I2P (Invoice-to-Pay)

SOFT SKILLS

Agile Methodology

Process Consultation

Problem-Solving

Cross-Functional Communication

LANGUAGES

Hindi: Native

English: C1 (Professional)

German: A2 (Basic)

WORK EXPERIENCE

Master Thesis

03/2025 - 07/2025

Bosch

Stuttgart, Germany

- Automated benchmarking of 10+ Large Language Models (LLMs) for BPMN activity mapping and conformance checking using a modular Python tool, streamlining integration across 7 REST APIs.
- Benchmarked LLM mapping accuracy against manually validated gold standards, delivering advanced evaluation metrics (precision, recall, F1) and achieving 20% improvement in workflow processing speed.
- Enabled end-to-end process mining conformance analysis by linking LLM-generated mappings with token-based replay metrics, supporting dynamic dashboards and business-ready validation.

Data Engineering in Process Mining Intern

09/2024 - 02/2025

Bosch

Stuttgart, Germany

- Streamlined Celonis platform tasks (Data Pool backups, job health checks) with 60% faster execution, and automated regex-based outlier detection for access anomalies, cutting manual audit effort by 80%.
- Designed Streamlit application which automated process conformance checks across Signavio and Celonis; enabled structured activity mapping workflows used in 3+ pilot use cases during the PoC phase.
- Launched a Gamified leader board using LLM-scored commits to boost versioning, increasing developer engagement by 45% and automated weekly adoption tracking via Celonis ML Workbench and Action Flow.

Working Student

11/2023 - 08/2024

Bosch

Stuttgart, Germany

- Optimized Celonis transformation queries and data model loading, cutting use case runtime and memory usage by 40%.
- Built automated permission reporting Views from the Knowledge Model, enabling quick detection of false permissions and improving reporting accuracy by 75%.

- Led a 6-day Celonis platform training for cross-functional participants (3 IT, 2 management), ensuring hands-on adoption of process mining best practices.

Senior Associate Consultant

 **Bosch Global Software Solutions**

 07/2022 – 08/2023

 Bangalore, India

- Led B2C process performance tuning, reducing runtime from 24 to 10 hours for 32M data through strategic query and memory optimization.
- Won 1st place in internal hackathon, designing innovative P2P views using the Knowledge Model, improving process transparency.
- Conducted a 1-day workshop for India-Germany teams on performance optimization and script versioning best practices, enhancing code quality awareness.

Software Engineer

 **Bosch Global Software Solutions**

 07/2020 – 06/2022

 Bangalore, India

- Orchestrated process flow transition from Sales Order to Purchase Order, covering 3.5M records across 7 systems from a 13-month dataset.
- Developed reusable O2C task templates across 8 regions, reducing manual scripting effort by 30%.
- Migrated 3 on-premise use cases to IBC, coordinating with Celonis Support to ensure seamless transition and functionality.

Associate Software Engineer

 **Bosch Global Software Solutions**

 09/2019 – 06/2020

 Bangalore, India

- Supported Celonis AR use case setup, uploading a 30M-record data model from 7 systems via Oracle SQL to SAP HANA, enabling large-scale process analysis.

EDUCATION

M.Sc. in Data Analytics and Decision Science

 **RWTH Aachen University**

 10/2023 – 09/2025

 GPA - 1,5

CO₂ EMISSIONS FORECASTING FOR EU COUNTRIES

ML ensemble prediction of EU carbon emissions

 Repository

- Developed a machine learning pipeline to **predict national CO₂ emissions** across EU countries by extending the methodology from a [2024 Scientific Reports paper](#).
- Benchmarked ML models (GBM, SVR, LSTM, Ensemble) across 10-year horizons in a Dockerized pipeline with Grafana dashboard and PostgreSQL-backed experimentation.

AUTOMATIC CONFORMANCE CHECKING INSIGHTS

Process Mining

 Repository

- This project is a "**Command-line based software**" that provides automatic conformance checking insights for the Celonis data models, that allows users to choose which insights should be computed for the event logs.
- The output of the insights can be used to optimize existing process models, make conformance checking decisions and gain a better understanding of internal processes captured by the analyzed event log.

B.Tech. in Computer Science and Engineering

 **SOA University, India**

 07/2015 – 06/2019

 CGPA: 9.83/10 (GPA: 1,1)

VEHICLE SPEED DETECTION FROM THE CAPTURED VIDEO

Computer Vision

- As a final year project we developed a software which will be used to extract objects from live highway camera recordings, and it will help to detect vehicles which are running above the maximum speed limit given in respect of roads or highways